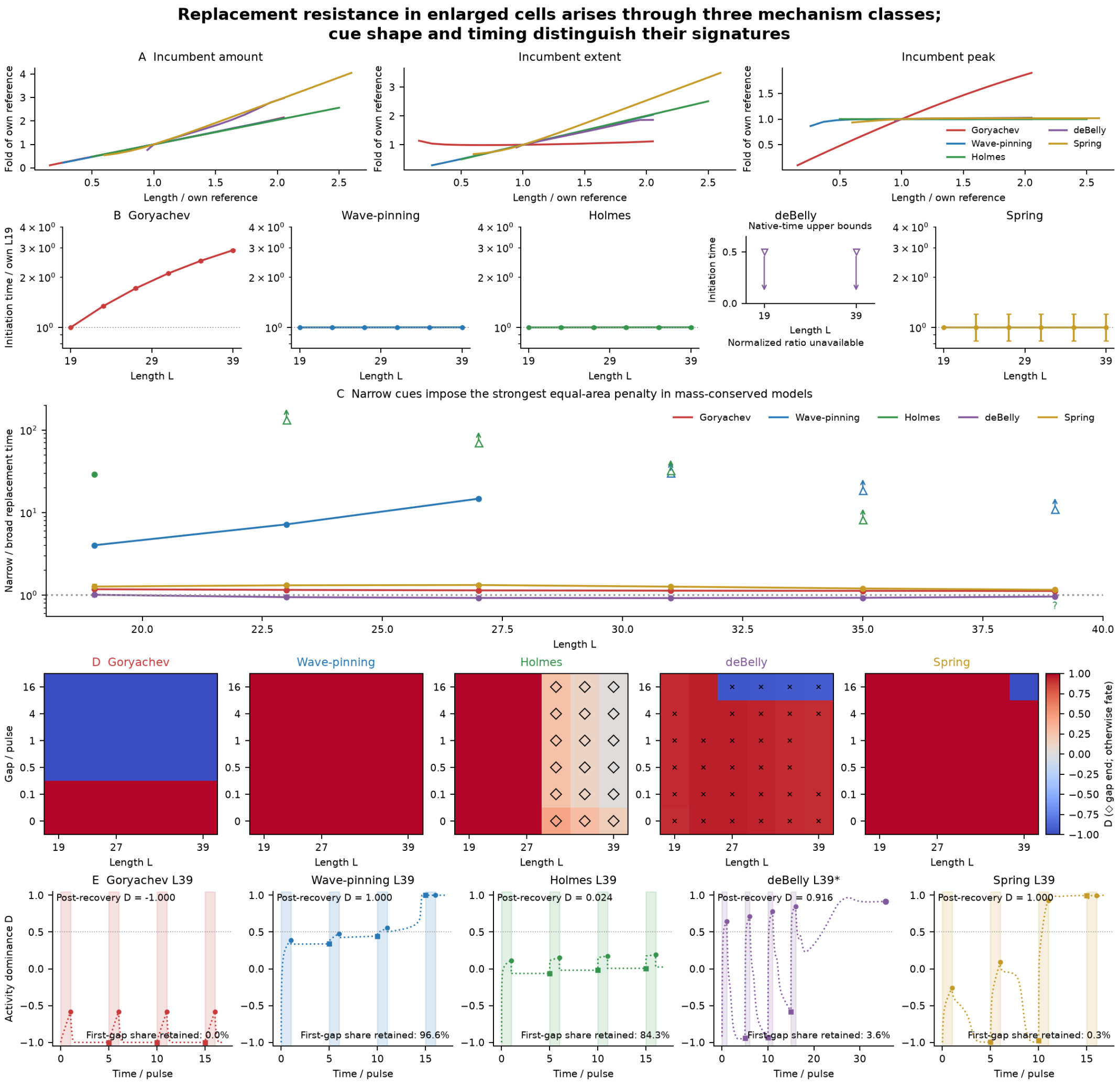
Figure 2. Shape and timing distinguish mechanism classes



Class / models	Initiation slows	Equal-area penalty	Share through gaps	Commit after collapse	Experimental measurement
Initiation-limited Goryachev	Yes (2.90×)	Modest (1.12×)	Resets	No	Beam width at fixed photon dose; pulse trains; gap-end biosensor share
Mass-conserved, saturating Wave-pinning, Holmes	No resolved slowing	Large; grows over resolved L range	Share retained (ratchet)	—	Beam width at fixed photon dose; pulse trains; gap-end biosensor share
Slow non-activity state deBelly*, Spring	Unresolved* / no	Modest (~1×)	Collapses early	Yes	Beam width at fixed photon dose; pulse trains; gap-end biosensor share

◇ Holmes L31–L39: last inter-pulse gap-end D, with no target-length calibration. × failed field tail; ? uninformative ratio.

* deBelly initiation is interval-censored; later-gap timing and endpoint fields carry numerical qualifications (complete supplement).

Figure 2 round2 with completed Holmes retention

Figure 2. Replacement resistance in enlarged cells arises through three mechanism classes; cue shape and timing distinguish their signatures

Distinguish replacement-resistance mechanisms using initiation, equal-area cue widths and measured pulse/gap retention.

Detailed legend

Enlarged cells resist replacement through distinct initiation, cue-width and gap-retention signatures, supporting three mechanism classes in the deterministic cue assays.

A, Incumbent amount, extent and peak across the retained 97 states. B, Initiation normalized to each model’s L19 value at central area and the fixed 10% L19 reporter threshold. Brackets propagate saved-sample intervals. deBelly L19 and L39 initiation both lie in (0, 0.5] native units; its normalized ratio is unavailable. Its separate native-time inset places open downward triangles at the measured upper bounds 0.5 and at the actual lengths 19 and 39, with arrows toward smaller times.

C, Narrow/broad replacement-time ratio at fixed commanded spatial area. Open triangles and arrows are one-sided bounds; question marks denote uninformative ratios. Wave-pinning has a growing large penalty. Holmes grows from about 29 at L19 to above 133 at L23; growth through L39 is unresolved because both width arms are censored. Goryachev and Spring have modest penalties rather than exact width invariance. Commanded area is not calibrated photon dose.

D, Four equal pulses with total ON 1.5 times the model-length commitment calibration; final D after recovery is the colour. For Holmes L31, L35 and L39, diamonds identify the last inter-pulse gap-end D from the targeted L27-referenced protocol; this endpoint is distinct from final fate. Gap0 uses the zero-duration boundary after the third pulse. No continuous commitment by 100,000 at these three lengths is extreme finite-horizon replacement resistance. Crosses preserve failed field tails, including 31 of the 36 deBelly fate cells.

E, L39 gap4 histories. Shading marks cue ON; circles and squares mark pulse and gap endpoints. The deBelly trace includes full recovery to its measured final fate; other traces show through one pulse duration of recovery, with full post-recovery D stated in each plot. Holmes retains more than 84% of raw challenger share through all three gaps, although D is slightly negative in the first two and final D remains below 0.5. Retention of share differs from the secondary S3 criterion requiring D at least zero. F, The class table has a numerical evidence row for every model and signature; membership is unchanged. deBelly early collapse and final replacement survive its separate quarter-step check, while later-gap timing and endpoint fields remain sensitive. All native kinetics are unchanged. These polarity readouts are distinct from observed migration and conditional OU persistence.

Figure 2. Replacement resistance in enlarged cells arises through three mechanism classes; cue shape and timing distinguish their signatures · continued

The separate Holmes L39 gap16 transfer has heterogeneous bounded input during gaps: S3 successes are 8/12, 9/12, 10/12 after gaps 1–3. Raw share ratios span 0.02152–1.78070; final D spans -0.063410–0.999944. Ten of twelve noisy histories replace, while matched-mean and zero controls do not. The class table describes gaps without added input; this separate transfer demonstrates variable retention and fate under spatially heterogeneous input.

Narrative points and conclusions

- Enlarged cells resist replacement through distinct initiation, cue-width and gap-retention signatures, supporting three mechanism classes in the deterministic cue assays.
- The targeted Holmes measurements report activity at every pulse and gap endpoint, independently of eventual replacement.
- No continuous commitment by 100,000 at Holmes L31, L35 and L39 is a finite-horizon finding of extreme replacement resistance.

Proposed measurements and experimental follow-up

Vary physical cue support at fixed commanded spatial input area and measure time to eligible75% challenger activity share.

Apply equal-total-ON pulse trains, measure activity share at every pulse and gap endpoint, then report postwithdrawal fate and field stability independently.

Scope, limitations and pending validation

deBelly normalized initiation unresolved within first-update intervals

Holmes width-penalty growth through L39 uninformative; moderate-length growth verified

deBelly later-gap timing and endpoint fields remain numerically sensitive

Holmes share retention is distinct from S3 D>=0 retention success and from commitment

Modest Goryachev/Spring width penalties qualify binary no-penalty wording; no class change

Selected mechanistic signatures do not causally allocate replacement resistance

Figure 2. Replacement resistance in enlarged cells arises through three mechanism classes; cue shape and timing distinguish their signatures · continued

Bounded heterogeneous Holmes gap input gives S3 successes 8/12, 9/12, 10/12; some histories lose share or replace, unlike deterministic/mean controls. Integrated gap exposure is 2.46 times ON exposure; deterministic class membership does not imply universal noisy-gap retention.

Sources and reproducibility

Class evidence CSV

Holmes endpoints

Figure 3. Separation slows the resolution of coexisting regions at short range; at matched separation, cell size dominates both driven takeover and resolution.

Separation slows the resolution of coexisting regions at short range; at matched separation, cell size dominates both driven takeover and resolution.

Figure 3. Separation slows the resolution of coexisting regions at short range; at matched separation, cell size dominates both driven takeover and resolution.

Separate matched-gap size effects from extra accessible separation for driven handover and cue-free resolution.

Detailed legend

Separation slows the resolution of coexisting regions at short range; at matched separation, cell size dominates both driven takeover and resolution. Times and gaps use native units.

A. A native cue is rotated at fixed cell size. The initial gap runs from the incumbent 10%-contrast edge to the cue support near edge. The inset holds the physical gap at 4 across small and large Wave-pinning rings; changing native widths also changes center separation. A1 measures continuous-cue complete-region handover. B withdraws the cue at complete-region challenger/incumbent excess ratio 1 and measures the first single resolved peak.

B. Left: Goryachev continuous-cue histories at circumference 37.8 and gaps 2 and 8; white dashed lines mark native continuous handover roots. Right: Wave-pinning at circumference 47.6, measured withdrawal gaps 2.78 and 14.84; the near condition resolves within 1140–1160, while the far condition remains unresolved through 20,000. Colour scales are shared within each model; the different assay time windows are shown explicitly. Native species are active chemical species. These maps display fields, not cell trajectories.

C. Goryachev handover rises with gap, whereas Holmes is nearly flat. Wave-pinning and Spring rise near the edge and then plateau; Wave-pinning does have a distance effect. Open circles are borderline placements. An × at the axis floor denotes an unavailable genuine two-region handover, not a time estimate. Size is consistently encoded by lightness and line style. The coloured top ticks mark initial antipodal-placement gaps.

D. Equal-amount cue-free resolution rises steeply at short range: approximately gaps 0–5 in Wave-pinning, 2–5.5 in Holmes, and 0.2–0.7 in Spring (the transition shifts with size). Thereafter observed curves flatten or remain unresolved. Holmes has 12/23 eligible ratio-1 main-curve times censored. The nine approved Holmes continuations to 80,000 resolve six histories; three middle-size histories remain unresolved at 80,000. Other Holmes lower bounds retain 20,000. These mixed horizons do not establish a downturn with gap. Unresolved times use open triangles and upward arrows; finite brackets use intervals, and lines do not bridge censored or ineligible placements. Top ticks mark each size's measured withdrawal gap in the initial-antipodal-cue trajectory; they do not assert exactly antipodal withdrawal centers. Goryachev includes a nonmonotone near-edge response. All alternate, merged and unseeded cases remain in the CSV and original native reader.

Figure 3. Separation slows the resolution of coexisting regions at short range; at matched separation, cell size dominates both driven takeover and resolution. · continued

E. Each larger size is compared with the smallest size at the largest shared measured gap, using piecewise-linear interpolation inside measured support. Solid bars represent size at matched gap; hatched bars represent extra separation to the larger-size antipodal-placement endpoint. Bars give $\log_{10}$ time ratios; finite B interval limits are marked. Open-ended bars retain one-sided bounds. A question mark means both relevant times are censored and the ratio is indeterminate. Such bars cannot be stacked into a measured total. Total ratios and all interpolation sources are tabulated separately. Size at matched gap dominates driven handover in all four models and the identifiable cue-free comparisons. Extra separation adds at most 1.15× for driven handover and about 1.01–1.07× in resolved Wave-pinning/Goryachev cue-free comparisons; Spring middle size is the supported exception (1.75×), with a larger matched-gap size term (4.29×). After extension, the Holmes middle-size matched-gap term is ≥2.37; its large-size term is only ≥0.618 and does not establish a positive size effect. Holmes extra-separation terms remain indeterminate. Enlarged cells' access to farther separations therefore does not generally explain their slower resolution.

The complete translation page, earlier main plate, all three B seeding ratios, A2 pulse brackets, 50%-edge reclassification, direct heading checks and numerical controls remain in the supplement and original reader. The finer-mesh Goryachev B ratio-1 gap-4 comparison has about 4% resolution-time sensitivity. Finite horizons establish lower bounds rather than permanent arrest; scalar/spatial fate and native-field stationarity are separate. The only new simulations are the nine approved Holmes cue-free continuations; no model retuning is included. Simulated activity heading, observed migration and conditional OU persistence remain distinct.

Narrative points and conclusions

- Separation slows the resolution of coexisting regions at short range; at matched separation, cell size dominates both driven takeover and resolution. Times and gaps use native units.
- At matched separation, size dominates both driven takeover and cue-free resolution. Extra separation contributes substantially in the supported Spring middle-size exception.
- The nine approved Holmes continuations extend the original exact endpoints to 80,000; unresolved conditions remain lower bounds and ratios of two censored times remain indeterminate.

Figure 3. Separation slows the resolution of coexisting regions at short range; at matched separation, cell size dominates both driven takeover and resolution. · continued

- The Holmes large-size matched-gap lower bound is below one and does not establish a positive large-size effect; the full qualification is retained in the legend and decomposition CSV.

Proposed measurements and experimental follow-up

Measure complete-region continuous-cue handover against initial edge gap at fixed circumference.

Withdraw at challenger/incumbent excess ratio 1 and measure first single-region resolution against the actual withdrawal gap.

Decompose size at matched gap and extra separation using interpolation only within measured support; propagate censoring and saved-sample brackets.

Scope, limitations and pending validation

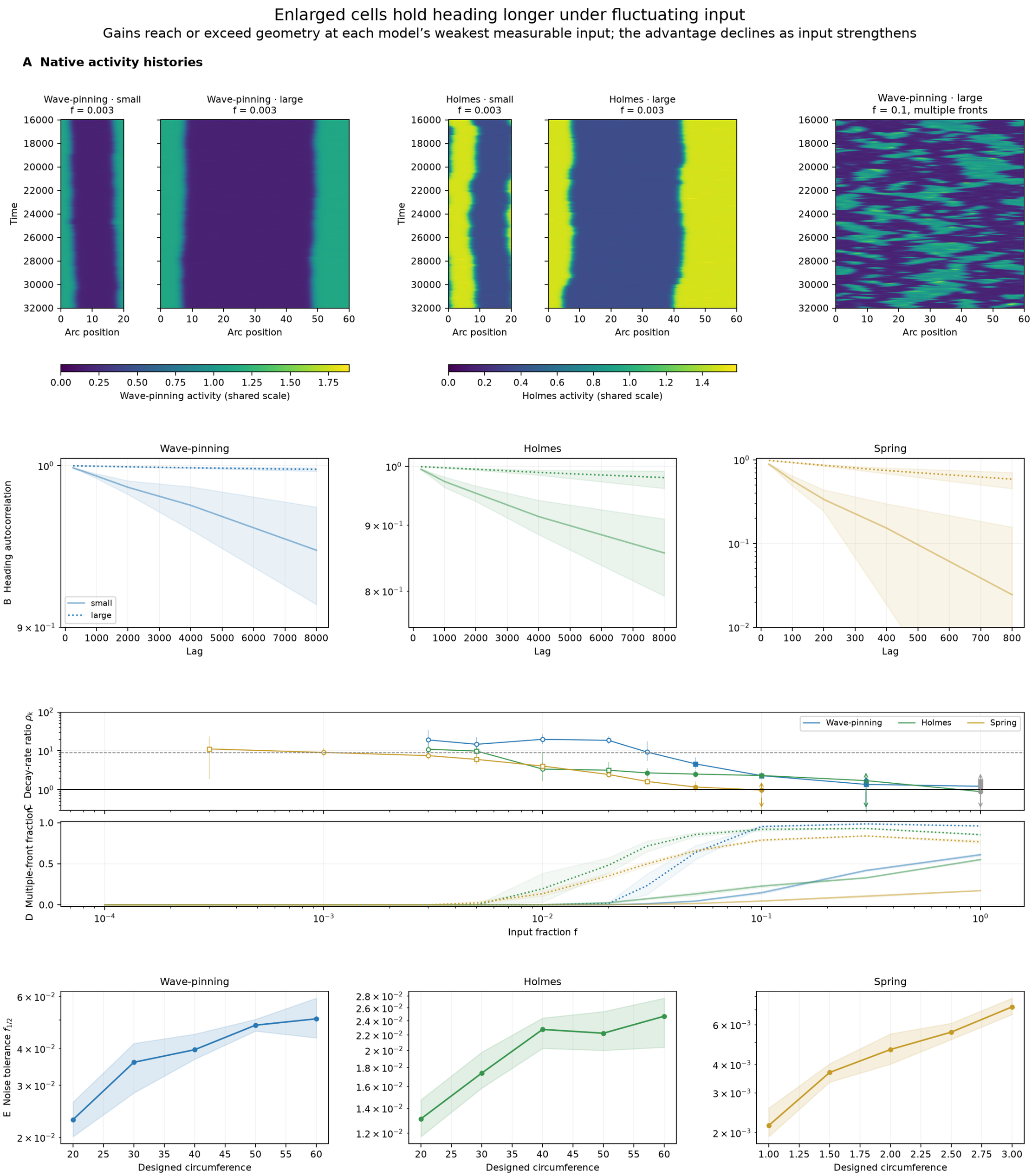
Censored ratios remain bounds or indeterminate; no horizon point substitution.

Nine approved Holmes extensions completed; three still censored at 80,000. Other cases retain 20,000. Holmes large-size matched-gap bound does not establish a positive size effect.

Sources and reproducibility

Complete corrected A1 evidence
All A2 switching brackets and unavailable calibrations
All B gap, fate, censoring and actual-heading measurements
Support-limited measured geometry remaps
Direct global-heading target checks
B numerical convergence including the material exception
Final artifacts and source-link audit
Lossless full native-history pixel audit
Current B evidence including all nine extensions
Matched-gap decomposition and censoring
Measured short-range evidence
Independent native-field and decomposition validation

Figure 4. Enlarged cells hold heading longer under fluctuating input



Enlarged cells hold heading longer under fluctuating input — at or above the geometric expectation at weak input — and the advantage declines as input strengthens.

Figure 4. Enlarged cells hold heading longer under fluctuating input — at or above the geometric expectation at weak input — and the advantage declines as input strengthens.

Measure the heading decay-rate advantage and describe geometric normalization and spatial organization under fluctuating input.

Detailed legend

Enlarged cells hold heading longer under fluctuating input — at or above the geometric expectation at weak input — and the advantage declines as input strengthens. The geometric statement describes model-specific weakest measurable point estimates: f = 0.003 for Wave-pinning and Holmes, and f = 0.0003–0.001 for Spring. Confidence intervals also admit sub-geometric gains in Holmes and Spring. At the shared displayed input f = 0.003, Spring is already below its actual geometric expectation (rate ratio 7.55 versus geometry 8.55), while its large-cell regime is still classified as single front. (A) Native activity kymographs for small and large Wave-pinning and Holmes at f = 0.003, plus a large Wave-pinning example at f = 0.1; width is proportional to circumference, with shared activity scales within each model. (B) Heading autocorrelation at f = 0.003 on model-specific logarithmic ordinates with whole-seed 95% bootstrap bands; intervals extending below zero continue beyond the positive plotting range. (C) Small/large rotational decay-rate ratio at the shortest usable lag satisfying the frozen lag rule, with pointwise 95% CIs. Reference lines show unity and nominal single-front geometry (9); actual-circumference normalization is reported in the supplement. Marker fill describes the large-cell regime; squares flag non-exponential lag dependence. (D) Multiple-front fractions with bootstrap intervals, reported descriptively: Holmes and Spring can fall below geometry while still single-front, so these data do not establish multiple fronts as the cause of erosion. (E) Fixed-reference-lag noise tolerance increases with size; smallest/largest CIs separate in all three shown models. Small cells are lighter solid, large cells darker dotted. Native units are used. Post hoc H1 amendment: evaluate gains and supported reversals only at inputs strictly below the point estimate of the largest cell’s own $f_{½}$; strict CI and size noise-tolerance rules are unchanged. Both main-view observation arms pass the amended H1 overall, while frozen rate H1 is inconclusive in both; all calls appear alongside in S0. H2/H3/A6 τ gates are retired. Goryachev is excluded from the main because its ratios derive from forced or non-exponential conditions, and remains complete in the supplement. Simulated polarity heading is distinct from migration persistence and conditional OU estimates.

Narrative points and conclusions

- Enlarged cells hold heading longer under fluctuating input — at or above the geometric expectation at weak input — and the advantage declines as input strengthens.

Figure 4. Enlarged cells hold heading longer under fluctuating input — at or above the geometric expectation at weak input — and the advantage declines as input strengthens. · continued

- Post hoc H1 passes in both main-view observation arms; frozen rate H1 remains inconclusive. All seven views, both arms and both forcing choices remain alongside in S0.
- Geometry and multiple-front fractions are descriptive. The data do not establish multiple fronts as the cause of the declining advantage.

Proposed measurements and experimental follow-up

Use endpoint coverage R ≥ 0.1 and a whole-seed bootstrap. Select the shortest usable lag at which small-cell autocorrelation is at most 0.98; report every usable lag and flag drift greater than 1.5-fold.

Measure fixed-reference-lag noise tolerance at autocorrelation 0.5 across every designed input, using log-input interpolation and explicit bounds.

Normalize the rate ratio by actual mean-circumference geometry, relative to the single-front expectation; this is not site diffusivity.

Scope, limitations and pending validation

The H1 domain restriction to inputs below the largest size's estimated noise tolerance is post hoc.

Weak-input geometric comparisons describe point estimates; Holmes and Spring confidence intervals also admit sub-geometric gains.

Non-exponential lags, forcing, off-grid lags and nonpositive autocorrelations remain explicit. Goryachev is complete in the supplement.

H2, H3 and A6 tau-based gates are retired; their frozen historical calls are preserved for provenance.

Simulated activity-heading persistence is distinct from observed migration and conditional OU persistence estimates.

Model definitions and readouts

$$C(\ell) = \langle \cos[\theta(t+\ell) - \theta(t)] \rangle, \qquad k(\ell) = -\frac{\ln C(\ell)}{\ell} \tag{1}$$

Figure 4. Enlarged cells hold heading longer under fluctuating input — at or above the geometric expectation at weak input — and the advantage declines as input strengthens. · continued

Both endpoints must satisfy the declared magnitude rule. The logarithmic rate is defined only for positive autocorrelation.

$$\rho_k = \frac{k_{\text{small}}}{k_{\text{large}}}, \qquad \rho_{\text{geo}} = \frac{\rho_k}{(\mathcal{C}_{\text{large}}/\mathcal{C}_{\text{small}})^2} \tag{2}$$

Calligraphic C denotes actual mean circumference. Geometric normalization is a relative single-front expectation.

$$C(\ell_{\text{ref}}; f_{1/2}) = \tfrac{1}{2} \tag{3}$$

The fixed reference lag is 2,000 for chemical models and 200 for Spring; crossing interpolation is logarithmic in input fraction.

Sources and reproducibility

E0
E1
E2
E3
E4
A1
A2
A3
A4
A5
A6
A7
windows_and_coverage
common_horizon
long_horizon
framing
independent_final_decision_audit
final_statistics_producer
S11_full_atlas_coverage
reviewed_interpretation_memo
reviewed_interpretation_memo_display
main_display_audit

Figure 4. Enlarged cells hold heading longer under fluctuating input — at or above the geometric expectation at weak input — and the advantage declines as input strengthens. · continued

Historical Spring observation-cadence qualification
Independent Figure 4 completion review
Frozen and post hoc H1 decisions
Every lag and lag-consistency flags
Complete lag-by-lag ratio supplement
Source and rendering receipt